Steady stream-power landscapes: existence, constructive local uniqueness, and singular limits

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Abstract

We study steady stream-power erosion coupled to diffusion and a specific-contributing-area conservation law. For all positive erosion exponents, positive Hölder forcing and bounded smooth domains with zero boundary elevation, we prove small-erosion existence of a classical elevation and a continuous conservative flux. A damped characteristic representation handles critical points without a Morse hypothesis; its prescribed-potential solvability principle is classical, following Richter. Our main nonlinear refinement is constructive local uniqueness for constant forcing and $m + n \geq 1$, under a flow-growth margin satisfied by every ellipsoid in dimension at least two. We give a convergent nested iteration, parameter stability and continuum residual bounds. An exact first-order ellipse calculation and numerical refinement tests illustrate the nonradial branch. On balls, we prove unique radial steady states for all positive parameters and exponents, and global evolution with exponential contraction for monotone radial data when $n \geq 1$. A universal small-diffusion profile yields finite, logarithmic and power-divergent summit regimes, together with weighted relief-moment thresholds. Second outer corrections produce a distinct threshold for finite-diffusion moment corrections. These theorems concern the specified stream-power system; general nonradial evolution and arbitrary-erosion nonradial existence remain open.

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1 Model, solution convention, and scope

Let $\Omega \subset \mathbb{R}^d$ be a bounded connected domain. Elevation z, specific contributing area a, and downhill water flux q obey

$$z_t = D\Delta z - Ka^m|\nabla z|^n + U, \tag{1}$$

$$\operatorname{div} q = \rho, \qquad q = -a\frac{\nabla z}{|\nabla z|}, \qquad z|_{\partial\Omega} = 0. \tag{2}$$

Here $D > 0$, $K \geq 0$, $m, n > 0$, and the rainfall normalization is $\rho = 1$ unless stated otherwise. For $\rho = 1$, the variable a is specific contributing area, not total upstream drainage area. With variable ρ, it is the corresponding rainfall-weighted specific discharge in the chosen normalization; this is an explicit extension of the constant-rainfall model. Equations (1)–(2) are the continuum model studied by Bonetti et al. [5]; the conservation-law formulation of contributing area and its critical points is developed in [4].

The normalized direction in (2) is undefined where $\nabla z = 0$. Throughout, the primary object is the flux q: its divergence equation holds on the entire domain in distributions, without additional point sources. On the noncritical set the displayed direction identity must hold. The solutions constructed below have $q = 0$ and $a = 0$ on the critical set, so this convention does not conceal a singular source.

Definition 1.1 (Classical-elevation, conservative-flux steady state)**.** A steady solution is a pair $z \in C^{2,\alpha}(\overline{\Omega})$, $q \in C^{0,\alpha}(\overline{\Omega}; \mathbb{R}^d)$, with $a = |q| \geq 0$, for which the steady equation (1) holds pointwise, $z = 0$ on the boundary, $\operatorname{div} q = \rho$ in distributions, and $q = -a\nabla z/|\nabla z|$ wherever $\nabla z \neq 0$.

The distinction between models matters. Binard, Degond and Noble [2] prove short-time well-posedness for a coupled water–bed–sediment system under strictly positive water depth. Birnir and Landry [3] establish global weak existence for a different, transport-limited sediment equation with prescribed water depth. Neither result is an existence theorem for (1)–(2). Hooshyar et al. [11] study variational formulations; Anand et al. [1] investigate self-similarity and vanishing diffusion, including explicit one-dimensional cases. Litwin et al. [12] give an all-exponent dimensional scaling of the specific-area model and examine its channel–hillslope coupling. We do not claim that one-dimensional hillslope formulas, dimensional scales, the logarithmic $m = n = 1$ limit, or the general idea of drainage characteristics originate here.

The main candidate contributions concern the nonlinear coupling: general-domain small-erosion existence, a constructive locally unique branch under the additional assumptions stated below, and the quantitative radial theory. The underlying prescribed-potential solvability principle is classical: Richter already allows critical points and positive forcing. We provide the explicit Hölder estimates and distributional formulation required by the coupled construction, rather than claiming a new linear existence criterion. The proofs below are complete at the level of standard elliptic and parabolic analysis, whose classical ingredients are specified where used.

Comparison with coefficient reconstruction

The coefficient-reconstruction ancestry is directly relevant to the linear drainage step, and must be separated from solving for the unknown landscape itself. Table 1 records only conclusions supported by the sources inspected; an unavailable theorem is not treated as excluded prior art.

The hypotheses of these conclusions are separate. In particular, the general existence theorem does not inherit the stronger theorem's uniqueness conclusion.

The landscape-specific comparisons are also equation-level: Anand et al. [1, Section 4.1 and Appendix C] already analyze one-dimensional vanishing diffusion and explicit profiles; Litwin et al. [12, Eq. 17] give the relevant dimensional scale. Binard et al. [2, Theorem 1.1] instead treat a three-field positive-depth Cauchy problem on $\mathbb{R}^2$, under additional exponent and Sobolev assumptions. None of these inspected statements is the general-domain coupled steady theorem above. Priority for the full theorem package remains provisional. The supplied full Richter paper has now been compared: its Theorem 1 covers the prescribed-potential sufficiency principle, and its Theorem 2 (p. 219) gives a positive-source, zero-boundary inverse setting with no Cauchy data. Neither theorem solves the present nonlinear erosion coupling.

2 Steady existence on arbitrary smooth domains

2.1 The damped drainage resolvent

Write

$$a = b|\nabla z|, \qquad q = -b\nabla z. \tag{3}$$

Source and locator	Prescribed data and setting	Result and distinction
Briane [6], Theorem 2.1 and Remark 2.2	Source-free $\operatorname{div}(\sigma\nabla u) = 0$; prescribed u; local nondegenerate saddle in $d = 2, 3$, with additional trajectory hypotheses	Bounded positive conductivity and reciprocal via an integrated-Laplacian condition; no positive-source coupled erosion equation.
Briane [7], Theorem 2.1	$u \in W^{2,1}_{\mathrm{loc}}$; uniformly continuous nonvanishing gradient; bounded Laplacian on $\mathbb{R}^d$	Positive locally bounded conductivity and reciprocal for the source-free equation; rough-field reconstruction, not the coupled problem.
Richter [8], Theorem 1, pp. 216–217	Prescribed u; $\lvert\nabla u\rvert \geq k_1$ or $\Delta u \geq k_2 > 0$; bounded characteristic solutions	Existence, uniqueness and sup bounds with inflow data. With $u = -z$ and empty inflow this already covers the linear sufficiency principle used here.
Theorem 2.2 here	Positive source; bounded smooth domain; inward uphill flow; arbitrary compact critical set	Explicit Hölder and distributional refinement of the classical solvability condition; necessity in the continuous nonnegative multiplier class is included.
Theorem 2.3 here	Same flux convention; unknown z coupled to erosion; positive forcing and zero elevation on the entire boundary	Small-K existence via compact fixed point, with Hölder flux and classical elevation; no elevation uniqueness claim.
Theorem 2.5 here	Unknown coupled landscape; constant positive forcing; $m + n \geq 1$; torsion-flow margin $L_0 < U/D$	Locally unique small-K branch in the bounded continuous multiplier class, with a convergent construction and continuum error bounds.

Table 1: A bounded comparison of mathematical settings, not an exhaustive novelty certificate.

The area equation becomes the nonsingular transport equation

$$v \cdot \nabla b + \delta b = \rho, \qquad v = -\nabla z, \quad \delta = \operatorname{div} v = -\Delta z. \tag{4}$$

No division by $|\nabla z|$ occurs in this equation.

Lemma 2.1 (Resolvent with critical points)**.** *Suppose $z \in C^{2,\alpha}(\overline{\Omega})$, $\partial_\nu z < 0$ on $\partial\Omega$, and*

$$0 < \kappa \leq \delta = -\Delta z \leq A.$$

Let L be a Lipschitz constant for ∇z on $\overline{\Omega}$, and assume $\alpha L < \kappa$. Let $0 < \rho_- \leq \rho \leq \rho_+$ with $\rho \in C^{0,\alpha}$. Define the backward downhill flow by

$$\partial_t X(t,x) = \nabla z(X(t,x)), \qquad X(0,x) = x,$$

and set

$$b_z(x) = \int_0^\infty \exp\Big[-\int_0^t \delta(X(s,x))\,\mathrm{d}s\Big]\,\rho(X(t,x))\,\mathrm{d}t. \tag{5}$$

Then $b_z \in C^{0,\alpha}$, $\rho_-/A \leq b_z \leq \rho_+/\kappa$, and

$$[b_z]_\alpha \leq \frac{[\rho]_\alpha}{\kappa - \alpha L} + \frac{\rho_+ [\delta]_\alpha}{\kappa(\kappa - \alpha L)}. \tag{6}$$

The flux $-b_z\nabla z$ has divergence ρ in distributions. This is the unique bounded continuous solution of (4) along characteristics. At every critical point x_*,

$$b_z(x_*) = \rho(x_*)/\delta(x_*). \tag{7}$$

Regime	Additional hypotheses	Conclusion
General-domain existence, Theorem 2.3	Positive Hölder forcing; $m, n > 0$; sufficiently small K	Classical elevation and conservative Hölder flux; uniqueness not asserted.
Local uniqueness, Theorem 2.5	Constant forcing; $m+n \geq 1$; $L_0 < U/D$; small K and a fixed torsion neighborhood	Unique branch in the stated bounded-multiplier class.
Radial steadiness, Theorem 4.1	Ball; constant forcing and unit rainfall; $m, n > 0$	Unique regular radial steady state for every positive parameter value.
Radial dynamics, Theorem 5.1	Same radial setting; compatible monotone initial data; $n \geq 1$	Global evolution and exponential radial contraction.

Table 2: Four regimes for the coupled model; uniqueness and dynamics are restricted to the classes displayed.

Proof. The vector field ∇z points strictly inward at the boundary. Its flow therefore exists in $\overline{\Omega}$ for every $t \geq 0$. There is no need to assume that each trajectory converges to a unique maximum. The exponential in (5) lies between e^{-At} and $e^{-\kappa t}$, which proves convergence and the two pointwise bounds.

Gronwall's inequality gives $|X(t,x) - X(t,y)| \leq e^{Lt}|x-y|$. The difference of the two integrands is bounded by the sum of

$$e^{-\kappa t}[\rho]_\alpha e^{\alpha L t}|x-y|^\alpha$$

and

$$\rho_+ e^{-\kappa t}[\delta]_\alpha \int_0^t e^{\alpha L s}\,\mathrm{d}s\,|x-y|^\alpha.$$

Integrating, and reversing the order of the latter integral, proves (6).

For completeness, the distributional equation follows without assuming spatial differentiability of b_z. Put

$$T_t f(x) = \exp\Big[-\int_0^t \delta(X(s,x))\,\mathrm{d}s\Big]\, f(X(t,x)).$$

This is the characteristic semigroup for the operator $-(v\cdot\nabla+\delta) = -\operatorname{div}(v\,\cdot)$. The exponential weight is the Jacobian determinant of the backward flow. For smooth coefficients the generator identity follows by differentiation and change of variables against a compactly supported test function. Approximation of the C^1 vector field in C^1, followed by uniform convergence of the flow and its Jacobian on finite time intervals, gives the same distributional identity here. Continuous f is then obtained by uniform approximation. Thus, for $b_T = \int_0^T T_t\rho\,\mathrm{d}t$,

$$\operatorname{div}(v b_T) = \rho - T_T\rho$$

in distributions. Uniform convergence $b_T \to b_z$ and $T_T\rho \to 0$ gives the assertion. Equivalently, integrating (4) along a characteristic gives the same formula with a terminal term $T_T b(x)$, of absolute value at most $e^{-\kappa T}\|b\|_\infty$; its limit is zero. This proves uniqueness in the stated characteristic class. At a critical point the flow is constant, and (7) is immediate. □

2.2 A sharper criterion for a prescribed landscape

The strict superharmonicity in Lemma 2.1 is convenient for the nonlinear fixed point, but stronger than necessary for the linear drainage problem.

Richter's positive-source reconstruction theory [8] is the closest antecedent, not only the source-free conductivity literature. Set $u = -z$. His equation $L(b;u) = \nabla b\cdot\nabla u + b\Delta u = \rho$ is

exactly (4). Under our compactness and boundary assumptions, positivity of Δu at critical points is equivalent to the existence of $k_1, k_2 > 0$ such that $|\nabla u| \geq k_1$ or $\Delta u \geq k_2$ everywhere. This is the partition hypothesis of [8, Theorem 1, pp. 216–217]. Moreover $\partial_\nu u > 0$, so the inflow boundary is empty. Richter defines bounded solutions along characteristics (p. 213) and proves existence, uniqueness and sup estimates. Thus the sufficiency principle below is not new. Our self-contained proof records a quantitative Hölder estimate, proves conservation distributionally across critical sets, and establishes necessity for a continuous nonnegative multiplier. These are the versions used by the coupled fixed point. Briane's source-free reconstructions [6, 7] provide related but different regularity theory.

Theorem 2.2 (Classical critical-set solvability with a Hölder refinement)**.** *Let Ω be a bounded C^{2,α_0} domain. Let $z \in C^{2,\alpha_0}(\overline{\Omega})$ have zero boundary values and $\partial_\nu z < 0$ on the boundary. Let $\rho \in C^{0,\alpha_0}$ be strictly positive. A bounded nonnegative continuous multiplier b with*

$$\operatorname{div}(-b\nabla z) = \rho$$

exists if and only if

$$-\Delta z(x) > 0 \qquad \text{at every } x \text{ with } \nabla z(x) = 0. \tag{8}$$

When this holds, the resolvent (5) *constructs such a multiplier, which is strictly positive and $C^{0,\alpha}$ for some $\alpha > 0$. It is unique among bounded continuous characteristic solutions. The assertion concerns the multiplier class $q = -b\nabla z$ with bounded continuous b, not every possible continuous contributing-area flux.*

Proof. For necessity fix a critical point x_*, write $H = D^2 z(x_*)$, and rescale

$$q_r(y) = \frac{-b(x_* + ry)\nabla z(x_* + ry)}{r}.$$

Continuity of b and differentiability of ∇z give $q_r \to -b(x_*)Hy$ uniformly on each bounded set. Meanwhile $\div_y q_r = \rho(x_* + ry)$ in distributions. Passing to the limit gives

$$\rho(x_*) = -b(x_*)\operatorname{tr} H = b(x_*)\delta(x_*).$$

The positive left side and nonnegative finite $b(x_*)$ force $\delta(x_*) > 0$.

For sufficiency, the critical set is compact and lies away from the boundary. By (8), choose $\kappa, \varepsilon > 0$ such that $\delta \geq \kappa$ wherever $|\nabla z| \leq \varepsilon$. Let $|\delta| \leq A$ globally. Along the backward flow,

$$\frac{\mathrm{d}}{\mathrm{d}t} z(X(t,x)) = |\nabla z(X(t,x))|^2.$$

Therefore the total time a trajectory can spend where $|\nabla z| > \varepsilon$ is at most $\operatorname{osc}(z)/\varepsilon^2$, uniformly in its starting point and its time horizon. Consequently

$$\int_0^t \delta(X(s,x))\,\mathrm{d}s \geq \kappa t - C_*, \qquad C_* = (A + \kappa)\operatorname{osc}(z)/\varepsilon^2. \tag{9}$$

The integrand in (5) is thus bounded by $\rho_+ e^{C_*} e^{-\kappa t}$. It follows that b is finite and strictly positive, with

$$\rho_-/A \leq b \leq \rho_+ e^{C_*}/\kappa.$$

Choose $\alpha < \min(\alpha_0, \kappa/L)$. The same difference estimate as in Lemma 2.1, now with the prefactor e^{C_*}, proves Hölder continuity. Indeed both exponential weights are at most $e^{C_*}e^{-\kappa t}$, and their difference is bounded by that quantity times the difference of their integrated exponents. The semigroup proof gives the distributional equation. The terminal term of any bounded characteristic solution tends to zero by (9), proving the stated uniqueness. □

The criterion allows regions where $\Delta z > 0$, provided the Laplacian has the required sign at every critical point. At a steady solution of the erosion equation that sign is automatic, since $-D\Delta z = U$ at a critical point. This observation does not supply the missing a priori estimates for an arbitrary-K nonlinear fixed point. It does explain why the drainage construction itself is not restricted to globally superharmonic topographies. For example, a flat maximum can admit a continuous flux while requiring an unbounded multiplier; such a case lies outside the criterion's stated class.

Two examples that separate the hypotheses. On $(-1, 1)$ with $\rho = 1$, let

$$z = 2 - 3x^2 + x^4, \qquad b = (6 - 4x^2)^{-1}, \qquad q = x.$$

Then $z(\pm 1) = 0$, its outward derivative is negative, and $-bz' = q$. The only interior critical point is zero, where $-z'' = 6 > 0$; near the boundary $-z'' = 6 - 12x^2 < 0$. Thus the critical-set theorem applies although global superharmonicity fails. In contrast, $z = 1 - x^4$ and $q = x$ also give continuous downhill drainage with $q' = 1$, but $b = 1/(4x^2)$ off zero is unbounded. This is outside the multiplier theorem, not a counterexample to it. Neither example is asserted to solve the coupled steady erosion equation with positive constant uplift.

2.3 A compact fixed point

Theorem 2.3 (General-domain steady existence)**.** *Let Ω be bounded, connected and C^{2,α_0}, $0 < \alpha_0 < 1$. Suppose $D > 0$, $m, n > 0$, and $U, \rho \in C^{0,\alpha_0}(\overline{\Omega})$ have strictly positive lower bounds. There exist $K_0 > 0$ and $\alpha \in (0, \alpha_0)$ such that every $K \in [0, K_0]$ admits a steady solution in the sense of Definition 1.1. In fact*

$$q = -b\nabla z, \quad a = b|\nabla z|, \quad b \in C^{0,\alpha}, \quad 0 < b_- \le b \le b_+.$$

If z_0 solves $-D\Delta z_0 = U$, $z_0|_{\partial\Omega} = 0$, then K_0 may be chosen so that

$$\tfrac{1}{2} z_0 \le z \le z_0, \qquad \|z - z_0\|_{C^{2,\alpha}} \le CK. \tag{10}$$

At every critical point of z,

$$b(x_*) = \frac{D\rho(x_*)}{U(x_*)}, \qquad \mathrm{D}_x q(x_*) = -\frac{D\rho(x_*)}{U(x_*)}\, D^2 z(x_*). \tag{11}$$

No convexity, radiality, or nondegeneracy of critical points is assumed. Uniqueness of z is not asserted.

Proof. The classical Dirichlet Poisson theorem, maximum principle and Hopf lemma give $z_0 \in C^{2,\alpha_0}$, $z_0 > 0$ in Ω, and $\partial_\nu z_0 \le -\eta < 0$ on its compact boundary; see [9, Chapters 3 and 6, in particular Theorem 6.14 for the Dirichlet problem]. These are the only domain-regularity facts needed.

Choose a sufficiently small C^2 neighborhood of z_0 in the zero-boundary affine space. Every function in it satisfies

$$-\Delta z \ge \kappa := \frac{\min U}{2D}, \qquad -\Delta z \le A, \qquad \partial_\nu z \le -\eta/2,$$

with common constants A and L bounding $-\Delta z$ and the Lipschitz constant of ∇z. A fixed extension operator on the smooth domain supplies a common global Lipschitz bound when Ω is nonconvex. Choose

$$0 < \alpha < \min\{\alpha_0, 1, m + n, \kappa/(2L)\}.$$

If $L = 0$ the last restriction is omitted. Fix a sufficiently small $\varepsilon > 0$ and put

$$\mathcal{C} = \{z : z|_{\partial\Omega} = 0, \quad \|z - z_0\|_{C^{2,\alpha}} \le \varepsilon\}.$$

Regarded as a subset of $C^{2,\beta}$ for a fixed $0 < \beta < \alpha$, this set is convex and compact. It is closed in that topology because the α-Hölder seminorm is lower semicontinuous under uniform convergence of second derivatives.

For each $z \in \mathcal{C}$ construct b_z by Lemma 2.1, and let

$$g_z = b_z^m |\nabla z|^{m+n}.$$

The positive upper and lower bounds on b_z, its uniform $C^{0,\alpha}$ norm, and $\alpha < \min(1, m+n)$ imply a common bound $\|g_z\|_{C^{0,\alpha}} \le G$. Let $P = (-\Delta_D)^{-1}$ be the zero-boundary Poisson inverse, with Schauder bound $\|Pf\|_{C^{2,\alpha}} \le C_P \|f\|_{C^{0,\alpha}}$. Define

$$\mathcal{F}_K(z) = z_0 - \frac{K}{D} P g_z. \tag{12}$$

For $K \le D\varepsilon/(C_P G)$ this map preserves $\mathcal{C}$.

We check the continuity needed for the fixed-point theorem rather than assume differentiability of the drainage map. If $z_j \to z$ in $C^{2,\beta}$ inside $\mathcal{C}$, their flows converge uniformly on finite time intervals. The integrands in (5) therefore converge uniformly there; their tails are uniformly exponentially small. Hence $b_{z_j} \to b_z$ uniformly. Uniform $C^{0,\alpha}$ bounds improve this to convergence in $C^{0,\beta}$. The same argument applies to g_{z_j}, including noninteger powers at $\nabla z = 0$. The Poisson estimate at exponent β proves continuity of $\mathcal{F}_K$ in $C^{2,\beta}$. Schauder's compact-convex fixed-point theorem now gives $z = \mathcal{F}_K(z)$.

The fixed-point equation is exactly

$$-D\Delta z + K b_z^m |\nabla z|^{m+n} = U.$$

Lemma 2.1 gives the global flux equation. Restricting K_0 further to $K_0 G \le \min U/2$ gives $U/2 \le -D\Delta z \le U$; comparison with the Poisson solution proves (10). The norm estimate follows from (12). At a critical point erosion vanishes, so $\delta = U/D$ and (7) proves the first identity in (11). Since b is continuous and $\nabla z(x_*) = 0$, Taylor expansion of ∇z proves the derivative formula for q. This derivative exists at the critical point even though only Hölder regularity is asserted for the flux globally. □

Remark 2.4 (What smallness means). The proof gives a positive threshold in terms of a Poisson Schauder constant, a flow Lipschitz bound, and the displayed Hölder estimates. It is an existence threshold, not a calibrated numerical channelization threshold. On a fixed rescaled domain with constant forcing, the natural erosion parameter is $KU^{n-1}R_{\mathrm{ref}}^{m+n}/D^n$, where R_{ref} is the chosen length scale. No claim about all values of this parameter on arbitrary domains is made.

2.4 A constructive locally unique nonradial branch

Constant forcing permits a different resolvent that removes the Laplacian from the multiplier's dependence on the trial elevation. This is useful because the original resolvent gives continuity but need not give a contraction in the norms used by the Poisson inverse.

Theorem 2.5 (Local uniqueness and convergent construction)**.** *Let $D, U, \rho > 0$ be constants, $m, n > 0$, $\sigma = m + n \ge 1$, and let Ω be a bounded connected $C^{2,\alpha}$ domain, $0 < \alpha < 1$. Put $\lambda = U/D$, $b_\circ = \rho/\lambda$, and let $-\Delta z_0 = \lambda$ with $z_0|_{\partial\Omega} = 0$. Suppose ∇z_0 has a global Euclidean*

Lipschitz constant $L_0 < \lambda$ on $\overline{\Omega}$. There exist $\varepsilon, K_* > 0$ such that for $0 \le K \le K_*$ there is exactly one steady state in the class

$$\|z - z_0\|_{C^{2,\alpha}} \le \varepsilon, \qquad q = -b\nabla z, \qquad b \text{ bounded, nonnegative and continuous.}$$

Here the boundary and distributional conventions are those of Definition 1.1. The multiplier is Lipschitz, $b_\circ \le b \le 2b_\circ$, with $\mathrm{Lip}(b) = O(K)$. A nested multiplier/Poisson iteration converges to this solution, with outer contraction in C^1. The solution depends Lipschitz-continuously on K in C^1 for z and C^0 for b. This is local uniqueness in the stated multiplier class, not uniqueness among arbitrary conservative fluxes or at arbitrary erosion intensity.

Proof. Write $\tau = K/D$ and $B = 2b_\circ$. Choose a small closed zero-boundary $C^{2,\alpha}$ ball $\mathcal{C}$ about z_0 such that, uniformly on it,

$$-\Delta z \ge \lambda/2, \quad \partial_\nu z < 0, \quad \mathrm{Lip}(\nabla z) \le L < \lambda, \quad \|\nabla z\|_\infty \le M.$$

A bounded extension operator controls global Lipschitz differences on nonconvex domains, so the strict L_0 margin persists in a sufficiently small ball. The set $\mathcal{C}$ is complete in the C^1 metric: its boundedness in $C^{2,\alpha}$ gives subsequential convergence in $C^{2,\beta}$, $\beta < \alpha$, and lower semicontinuity of the α seminorm puts the C^1 limit back in the ball. For $G_z = |\nabla z|^\sigma$ there are uniform constants

$$\|G_z\|_\infty \le G_0 = M^\sigma, \qquad \mathrm{Lip}(G_z) \le G_1 = \sigma M^{\sigma-1} L, \qquad \|G_z - G_w\|_\infty \le C_\sigma \|\nabla z - \nabla w\|_\infty,$$

where $C_\sigma = \sigma M^{\sigma-1}$. Enlarge M if necessary so $M > 0$. These estimates include $\sigma = 1$.

Multiplier contraction. For each $z \in \mathcal{C}$ let X_z be its uphill flow and define, on continuous $b_\circ \le b \le B$,

$$\mathcal{T}_{\tau,z} b(x) = b_\circ + \tau \int_0^\infty e^{-\lambda t} b(X_z(t,x))^{m+1} G_z(X_z(t,x))\,\mathrm{d}t. \tag{13}$$

Put $A = (m+1)B^m G_0$. The map preserves this interval if $\tau B^{m+1} G_0/\lambda \le b_\circ$, and contracts in the supremum norm with factor $\eta = \tau A/\lambda$. Impose $\eta \le 1/2$. Its unique fixed point, denoted $\widehat{b}_{\tau,z}$, is Lipschitz: iteration from the constant $b_\circ$, using flow separation e^{Lt}, gives

$$\mathrm{Lip}(\mathcal{T}_{\tau,z} b) \le \frac{\tau}{\lambda - L}\{A\,\mathrm{Lip}(b) + B^{m+1} G_1\}.$$

Imposing also $\tau A/(\lambda - L) \le 1/2$ gives

$$\mathrm{Lip}(\widehat{b}_{\tau,z}) \le \frac{2\tau B^{m+1} G_1}{\lambda - L}. \tag{14}$$

Uniform convergence of the iterates preserves this bound.

Dependence on elevation. For $e = \|\nabla z - \nabla w\|_\infty$, Gronwall gives $|X_z(t,x) - X_w(t,x)| \le e\int_0^t e^{L(t-s)}\,\mathrm{d}s$. Let H_* be a uniform Lipschitz bound for $\widehat{b}_{\tau,w}^{m+1} G_w$ on the selected small τ interval; (14) supplies one. Splitting the fixed-point difference into multiplier, gradient and trajectory differences yields

$$(1-\eta)\|\widehat{b}_{\tau,z} - \widehat{b}_{\tau,w}\|_\infty \le \tau\left\{\frac{B^{m+1} C_\sigma}{\lambda} + \frac{H_*}{\lambda(\lambda - L)}\right\} e.$$

Here the second denominator follows by integrating $e^{-\lambda t}\int_0^t e^{L(t-s)}\,\mathrm{d}s$ over $t > 0$. Consequently

$$\|\widehat{b}_{\tau,z} - \widehat{b}_{\tau,w}\|_\infty \le C_b \tau \|z - w\|_{C^1}. \tag{15}$$

Since $b \mapsto b^m$ is Lipschitz on $[b_\circ, B]$, including when $m < 1$, $g_{\tau,z} = \widehat{b}_{\tau,z}^m G_z$ satisfies $\|g_{\tau,z} - g_{\tau,w}\|_\infty \le C_g \|z - w\|_{C^1}$ with C_g uniform for small τ. It also has a uniform Lipschitz norm.

Poisson contraction. Define

$$\widehat{\mathcal{F}}_\tau(z) = z_0 - \tau P g_{\tau,z}. \tag{16}$$

Uniform $C^{0,\alpha}$ bounds on g and the Schauder estimate make this a self-map of $\mathcal{C}$ for small τ. The estimate $\|Pf\|_{C^1} \le C_E\|f\|_\infty$ follows from Dirichlet $W^{2,p}$ regularity with a finite $p > d$ and Sobolev embedding. Thus the map contracts in C^1 with $Q = \tau C_E C_g$; reduce τ so $Q \le 1/2$. Banach's theorem proves existence and convergence of the outer iteration on the complete metric space $\mathcal{C}$.

Equivalence and uniqueness. The multiplier identity implies, along characteristics,

$$-\nabla z \cdot \nabla b + \lambda b = \rho + \tau b^{m+1} G_z.$$

At the Poisson fixed point $-\Delta z = \lambda - \tau b^m G_z$. Combining these equations gives $\operatorname{div}(-b\nabla z) = \rho$ in distributions, and the Poisson equation is exactly the steady erosion equation. The constructed Lipschitz multiplier justifies this calculation almost everywhere.

Conversely, any pair in the stated class obeys the characteristic identity. Indeed, away from zeros of the C^1 vector field, a flow-box change of variables turns the distributional transport equation into a one-dimensional equation with continuous right side, including the flow Jacobian. Integration first for almost every transverse coordinate, then continuity, gives the identity on every trajectory. At zeros the critical-point rescaling argument gives $(-\Delta z)b = \rho$. The original resolvent and $-\Delta z \ge \lambda/2$ imply $b \le B$. The steady equation also gives $-\Delta z \le \lambda$, so $b \ge b_\circ$. Rewriting the transport equation with constant damping λ and letting its bounded terminal term tend to zero gives (13). Hence its multiplier and elevation are precisely the two unique fixed points.

Parameter dependence. For fixed z, subtraction of the multiplier fixed-point equations at τ, τ' and absorption of η give $\|\widehat{b}_{\tau,z} - \widehat{b}_{\tau',z}\|_\infty \le C|\tau - \tau'|$. The same bound holds for the Poisson maps in C^1. Absorbing their uniform contraction factor Q proves Lipschitz dependence of the fixed elevation; (15) then proves it for the multiplier. Since D is fixed, these are also Lipschitz bounds in K. □

Remark 2.6 (A checkable nonradial geometry). For the ellipsoid $\sum_i x_i^2/A_i^2 < 1$,

$$z_0 = \frac{\lambda(1 - \sum_i x_i^2/A_i^2)}{2\sum_i A_i^{-2}}, \qquad L_0 = \lambda \frac{\max_i A_i^{-2}}{\sum_i A_i^{-2}} < \lambda \quad (d \ge 2).$$

Thus all such ellipsoids satisfy the hypothesis. Elongation can make the margin small; no uniform threshold over all aspect ratios is claimed. For the ellipse $x^2/4 + y^2 < 1$ used below, $L_0/\lambda = 4/5$ and $(\lambda - L_0)/\lambda = 1/5$. These ratios verify the torsion-flow hypothesis only; they do not instantiate the invariant-ball radius, the Poisson constants or an admissible numerical K_*. On a convex domain the trajectory estimates and their integrability denominators can instead use a uniform one-sided flow-growth rate $\mu < \lambda$, while the spatial Lipschitz bounds for G_z retain an ordinary Lipschitz constant for ∇z. Such a growth rate follows, for example, from $D^2 z \le \mu I$ throughout the invariant ball. Strict concavity of z_0 therefore suffices, including on an interval. On a nonconvex domain that pointwise Hessian inequality alone does not give the required global pairwise trajectory estimate.

Proposition 2.7 (A consolidated sufficient smallness choice)**.** *Fix the invariant ball $\mathcal{C}$ of radius ε and its constants from Theorem 2.5. Let C_S satisfy $\|Pf\|_{C^{2,\alpha}} \le C_S\|f\|_{C^{0,\alpha}}$, and put*

$$\overline{\tau} = \min\left\{\frac{\lambda b_\circ}{B^{m+1}G_0}, \frac{\lambda}{2A}, \frac{\lambda - L}{2A}\right\}, \qquad M_m = m\max(b_\circ^{m-1}, B^{m-1}).$$

Define the following constants independently of $0 \le \tau \le \overline{\tau}$*:*

$$L_b = \frac{2\overline{\tau}B^{m+1}G_1}{\lambda - L}, \qquad H_* = AL_b + B^{m+1}G_1,$$
$$C_b = 2\left\{\frac{B^{m+1}C_\sigma}{\lambda} + \frac{H_*}{\lambda(\lambda - L)}\right\}, \qquad C_g = B^m C_\sigma + M_m G_0 C_b \overline{\tau},$$
$$J = M_m G_0 L_b + B^m G_1, \qquad G_\alpha = B^m G_0 + J\,\mathrm{diam}(\Omega)^{1-\alpha}.$$

Then a sufficient choice is

$$0 \le K \le D \min\left\{\overline{\tau}, \frac{\varepsilon}{C_S G_\alpha}, \frac{1}{2C_E C_g}\right\}.$$

Proof. The three bounds defining $\overline{\tau}$ give multiplier-interval invariance, $\eta \le 1/2$, and the Lipschitz-iteration bound, respectively. Equation (14) gives L_b; the elevation-difference estimate then gives the displayed C_b, C_g. The forcing has supremum norm at most $B^m G_0$ and Lipschitz constant at most J, hence Hölder norm at most G_α. The last two smallness bounds make the Poisson map preserve $\mathcal{C}$ and give $Q \le 1/2$. □

This consolidates sufficient analytical inequalities; it is not a numerical instantiation of their domain-dependent constants or a channelization threshold.

Corollary 2.8 (Continuum residual bounds)**.** *Use the constants* $\eta, Q < 1$ *and* C_b, C_E *from the preceding proof. For an approximate* $z \in \mathcal{C}$ *and continuous* $b_\circ \le b \le B$*, put*

$$e_b = \|b - \mathcal{T}_{\tau,z} b\|_\infty, \qquad e_z = \|z - z_0 + \tau P(b^m G_z)\|_{C^1}, \qquad M_m = m\max(b_\circ^{m-1}, B^{m-1}).$$

The unique exact pair (z_*, b_*) *satisfies*

$$\|z - z_*\|_{C^1} \le \frac{e_z + \tau C_E M_m G_0 e_b/(1-\eta)}{1 - Q}, \qquad \|b - b_*\|_\infty \le \frac{e_b}{1-\eta} + C_b \tau \|z - z_*\|_{C^1}.$$

Proof. The inner contraction gives $\|b - \widehat{b}_{\tau,z}\|_\infty \le e_b/(1-\eta)$. Replacing b by $\widehat{b}_{\tau,z}$ changes the Poisson term by at most $\tau C_E M_m G_0 e_b/(1-\eta)$. Apply the outer contraction residual estimate, then (15). □

These are continuum norms. A numerical certificate additionally requires bounds for discretization, flow integration, the infinite tail, and membership in the invariant set. Finite-grid residuals alone do not meet those requirements.

Corollary 2.9 (Inexact nesting and a finite inner stopping rule)**.** *Fix* $\tau > 0$ *in the preceding range and let* $\eta, Q < 1$ *be uniform contraction factors. Suppose each approximate elevation* z_j *belongs to* $\mathcal{C}$*. For its inner solve, use an interval-preserving approximation* $\widetilde{\mathcal{T}}_j$ *to* $\mathcal{T}_{\tau,z_j}$*, with error at most* δ_j *at every evaluated inner iterate. Given* $r_j > 0$*, require*

$$\delta_j \le (1-\eta) r_j / 4.$$

Starting from $b^0 = b_\circ$*, iterate* $b^{k+1} = \widetilde{\mathcal{T}}_j b^k$ *and stop when* $\|b^{k+1} - b^k\|_\infty \le r_j$*; retain* $b_j = b^k$*. This stopping condition is reached after finitely many steps. If*

$$\pi_j \ge \|z_{j+1} - z_0 + \tau P(b_j^m G_{z_j})\|_{C^1}, \qquad \xi_j = \pi_j + \frac{\tau C_E M_m G_0}{1-\eta}(r_j + \delta_j),$$

then, for $E_j = \|z_j - z_*\|_{C^1}$*,*

$$E_{j+1} \le Q E_j + \xi_j, \qquad E_j \le Q^j E_0 + \sum_{i=0}^{j-1} Q^{j-1-i}\xi_i.$$

In particular, $\xi_j \le C\theta^j$ *with* $0 < \theta < 1$ *gives convergence; when* $\theta = Q$ *the sum is at most* CjQ^{j-1}*.*

Proof. Writing $d_k = \|b^k - \widehat{b}_{\tau,z_j}\|_\infty$ gives $d_k \le \eta^k d_0 + \delta_j/(1-\eta)$. Hence

$$\|b^{k+1} - b^k\|_\infty \le (1+\eta)\eta^k d_0 + \frac{2\delta_j}{1-\eta} \le (1+\eta)\eta^k d_0 + r_j/2.$$

Here $d_0 \le B - b_\circ$, so any k with $(1+\eta)\eta^k(B-b_\circ) \le r_j/2$ guarantees the stopping test. At stopping, the exact inner residual is at most $r_j + \delta_j$. The inner contraction and the Poisson bound therefore give $\|z_{j+1} - \widehat{\mathcal{F}}_\tau(z_j)\|_{C^1} \le \xi_j$. The outer contraction proves the recurrence; induction proves its sum. For $\tau = 0$ the exact solution is already $z_0, b_\circ$. □

The inner error can be budgeted without conflating its sources. For a truncated flow integral at time T_j, let $e_{X,j}$ bound the error from approximate trajectories and $e_{I,j}$ the quadrature and integrand-representation error, both in the supremum norm of the multiplier map and including its factor τ. Then one may use

$$\delta_j = e_{X,j} + e_{I,j} + \frac{\tau B^{m+1} G_0}{\lambda} e^{-\lambda T_j}.$$

Pointwise clipping of the computed map to $[b_\circ, B]$ preserves this error bound, since the exact map takes values in that interval. The distinct error π_j covers the Poisson solve, spatial representation and any subsequent modification of the elevation. Thus geometrically decreasing r_j, δ_j and π_j, subject to the displayed inner-error inequality, give the stated convergent finite-tolerance construction. An implementation must bound the continuum norms used in its stopping test; sampled norms alone are insufficient. It must also keep every z_j in $\mathcal{C}$: small C^1 errors do not by themselves ensure membership of a $C^{2,\alpha}$ ball. Any admissibility correction must be accounted for in π_j. No such verified implementation or instantiated error budget is claimed for the collocation experiment below.

3 Response at zero erosion and a nonradial exact check

Corollary 3.1 (First-order elevation response)**.** *For any family of the solutions furnished by Theorem 2.3, as $K \downarrow 0$,*

$$z_K = z_0 - \frac{K}{D} P\big(b_0^m |\nabla z_0|^{m+n}\big) + o(K) \quad \text{in } C^{2,\beta}, \quad 0 < \beta < \alpha, \tag{17}$$

where b_0 is the resolvent (5) *for z_0. The first-order coefficient is independent of which small-K steady solution is selected.*

Proof. The norm estimate gives $z_K \to z_0$. The continuity proved above gives $g_{z_K} \to g_{z_0}$ in $C^{0,\beta}$. Divide (12) by K and use the continuous Poisson inverse. □

For $D = U = \rho = 1$ one has $-\Delta z_0 = 1$ and $b_0 = 1$. With $p_0 = m + n$, the spatial shape of the first correction is $P(|\nabla z_0|^{p_0})$. Moreover,

$$b_K = 1 + K b_1 + o(K), \qquad b_1(x) = \int_0^\infty e^{-t} |\nabla z_0(X_0(t,x))|^{p_0} \, \mathrm{d}t, \tag{18}$$

in $C^{0,\beta}$. Indeed, $\delta_K = 1 - K g_{z_K}$ in (5); expanding its exponential proves the formula uniformly. The bounds for $(b_K - 1)/K$ in $C^{0,\alpha}$ follow from (6) and $\delta_K - 1 = -K g_{z_K}$, and interpolation gives the stated topology.

As an exactly checkable nonradial example, take $\Omega = \{x^2/4 + y^2 < 1\}$, $D = U = \rho = 1$ and $m = n = 1$. Then

$$\begin{aligned} z_0 &= \frac{2}{5} - \frac{x^2}{10} - \frac{2y^2}{5}, \qquad g_0 = \frac{x^2 + 16y^2}{25}, \\ W &= \frac{(1 - x^2/4 - y^2)(44 + 3x^2 + 52y^2)}{1025}, \\ b_1 &= \frac{x^2}{35} + \frac{16y^2}{65}. \end{aligned} \tag{19}$$

Direct differentiation gives $-\Delta W = g_0$, $W|_{\partial\Omega} = 0$, and $-\nabla z_0 \cdot \nabla b_1 + b_1 = g_0$. Thus $z = z_0 - KW + o(K)$ and $b = 1 + Kb_1 + o(K)$. Substitution of the truncated pair into both coupled equations has zero coefficients at orders K^0 and K^1. The supplied script checks these rational polynomial identities exactly. This is not an exact finite-K landscape; the truncation has a generally nonzero K^2 residual.

Finite-erosion numerical refinement

For the same ellipse, write $x = 2u$, $y = v$ and approximate $z = (1 - u^2 - v^2)\sum_{i+j\le N} a_{ij}u^{2i}v^{2j}$ and $b = \sum_{i+j\le N} b_{ij}u^{2i}v^{2j}$. The boundary elevation vanishes exactly. An analytic-Jacobian least-squares solve enforces the two stationary equations on a collocation grid. Table 3 reports maximum absolute residuals on a separate staggered grid of 1,457 interior points, at $K = 0.2$. The supplied code also checks $K = 0.025, 0.05, 0.1$. The $N = 4$ summit is approximately

N	Bed residual	Area residual
2	1.49×10^{-5}	1.39×10^{-5}
3	1.17×10^{-7}	9.57×10^{-8}
4	7.83×10^{-10}	9.56×10^{-10}

Table 3: Independent-grid residual refinement for an approximate nonradial steady state. Values are rounded upward.

0.39155525, compared with the zero-erosion value 0.4. Its discrepancy from the exact first-order response on the test grid is 1.45×10^{-4}; at $K = 0.1$ it is 3.65×10^{-5}, consistent with a quadratic correction. Separately, treating the saved version 1.2 decimal coefficient strings as exact rationals, the coefficient ℓ^1 bound on $[-1, 1]^2$ proves whole-ellipse bounds for the $N = 4$, $K = 0.2$ polynomial:

$$\|R_{\mathrm{bed}}\|_\infty < 2.81 \times 10^{-6}, \quad \|R_{\mathrm{area}}\|_\infty < 2.76 \times 10^{-6}, \quad b > 0.9263, \quad -\Delta z > 0.8269.$$

The boundary factor of z is positive, and its boundary value is exactly zero. The bounds follow by summing absolute nonconstant coefficients, using $|u|, |v| \le 1$; the accompanying exact-rational checker reproduces them. They bound the specified polynomial residuals, not its error from a true solution or membership in the contraction ball.

These are floating-point, finite-grid diagnostics. They neither verify membership in the theorem's invariant ball nor show that the selected parameters lie below K_*. The solver is not the proved contraction iteration, and these residuals do not supply the continuum norms required by Corollary 2.8.

4 All-parameter radial steady states

Here $\Omega = B_R$, $U > 0$ is constant and $\rho = 1$. Write $r = |x|$ and seek a strictly downhill radial elevation, with slope $s = -z'(r) > 0$.

Theorem 4.1 (Radial existence, uniqueness and summit regularity)**.** *For every $d \geq 1$, $R, D, K, U > 0$, and $m, n > 0$, there is exactly one regular downhill radial steady solution without an interior point source. It has*

$$a(r) = r/d, \quad q(x) = x/d, \quad z(r) = \int_r^R s(t)\,\mathrm{d}t, \tag{20}$$

where s is the positive solution of

$$D\left(s' + \frac{d-1}{r}s\right) + \frac{K}{d^m} r^m s^n = U, \qquad s(0) = 0. \tag{21}$$

Put $A = U/(dD)$ and $\sigma = m + n$. Then

$$0 < s(r) \leq Ar, \qquad 0 \leq z(r) \leq \frac{U}{2dD}(R^2 - r^2), \tag{22}$$

$$s(r) = Ar - Br^{\sigma+1} + o(r^{\sigma+1}), \qquad B = \frac{KA^n}{Dd^m(d+m+n)}. \tag{23}$$

The elevation is strictly positive for $r < R$. In particular, $z \in C^{2,\gamma}(\overline{B}_R)$ for every $0 < \gamma < \min\{1, m+n\}$, and is smooth away from the summit. The steady slope and elevation increase with U and decrease with K.

Proof. The flux conservation law gives $(r^{d-1}a)' = r^{d-1}$. Its integration constant is zero: a nonzero constant is a point flux at the origin (or a nonzero central flux in the one-dimensional symmetric formulation). Hence $a = r/d$ and $q = x/d$. The elevation equation is (21).

Set $c_0 = K/(Dd^m)$ and $s = rw$. The integral equation is

$$w(r) = A - c_0 r^{-d} \int_0^r t^{d+m+n-1} w(t)^n\,\mathrm{d}t. \tag{24}$$

On a sufficiently short interval this is a contraction on continuous functions with $A/2 \leq w \leq 3A/2$. The power function is Lipschitz on this positive interval, even when $n < 1$, and the contraction factor is $O(r^{m+n})$. It follows that $s > 0$ near zero. A first zero at positive r is impossible because the ODE would give $s' = U/D > 0$. Integrating the ODE after multiplication by r^{d-1} proves $s \leq Ar$. Standard ODE continuation on intervals bounded away from zero now extends the solution to every finite radius.

For two positive regular solutions, $\delta_s = s_1 - s_2$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\tfrac{1}{2} r^{2(d-1)} \delta_s^2\right) = -c_0 r^{2(d-1)+m} \delta_s (s_1^n - s_2^n) \leq 0.$$

The expression tends to zero at the origin, proving uniqueness. Substitution of $w \to A$ in (24) gives (23). More explicitly, the integral equation and its differentiated ODE give $w - A = O(r^\sigma)$, $w' = O(r^{\sigma-1})$ and $w'' = O(r^{\sigma-2})$. The elevation Hessian is $-wI - (w'/r)xx^T$: its difference from $-AI$ is $O(r^\sigma)$ and its spatial derivative away from zero is $O(r^{\sigma-1})$. Splitting pairs of points according to whether their separation is smaller than their distance from zero proves the asserted $C^{2,\gamma}$ regularity for every $\gamma < \min(1, \sigma)$. Integrating the slope gives (22). Scalar first-crossing comparison in (21), using the leading term or (23) at the origin, proves parameter monotonicity. □

Remark 4.2 (No hidden radial minima). In the regular radial class with finite continuous area and no point flux, the summit Hessian is $-U/(dD)$ times the identity. If a downhill derivative first returned to zero at $r > 0$, the steady elevation equation there would force $z'' = -U/D < 0$, contradicting that first return. Thus regular radial states in this class are downhill; the theorem is not merely uniqueness under an arbitrary prescribed slope sign. It says nothing about uniqueness among nonradial states.

For $n = 1$ there is a useful closed quadrature:

$$s(r) = \frac{U}{D} \frac{\displaystyle\int_0^r t^{d-1} \exp\!\left(\frac{K t^{m+1}}{D d^m (m+1)} \right) \mathrm{d}t}{\displaystyle r^{d-1} \exp\!\left(\frac{K r^{m+1}}{D d^m (m+1)} \right)}. \tag{25}$$

The $d = 1$, $m = n = 1$ member is the familiar Dawson-function hillslope, not a new special case; compare [5, 1].

5 Global evolution and relaxation in the radial class

Theorem 5.1 (Global monotone radial evolution)**.** *Assume $m > 0$, $n \geq 1$, and $D, K, U, R > 0$. Let z_{in} be a nonincreasing radial $C^{2,\alpha}$ function on $\overline{B}_R$, with $z_{\mathrm{in}}(R) = 0$, regular center, and the classical compatibility condition*

$$D\left(z_{\mathrm{in}}''(R) + \frac{d-1}{R} z_{\mathrm{in}}'(R) \right) - K(R/d)^m |z_{\mathrm{in}}'(R)|^n + U = 0.$$

Here $\alpha > 0$ may be decreased below $\min(1, m)$. Then the radial equation

$$z_t = D\left(z_{rr} + \frac{d-1}{r} z_r \right) - k(r)|z_r|^n + U, \qquad k(r) = K(r/d)^m, \tag{26}$$

has a unique global solution, classical for positive time and continuous with the initial data. It is strictly downhill for $0 < r \leq R$, $t > 0$, and therefore solves the coupled system with $a = r/d$, $q = x/d$.

Let $s_{\mathrm{in}} = -z_{\mathrm{in}}'$, $c = \sup_{0<r\leq R} s_{\mathrm{in}}(r)/r$, and choose

$$\overline{U} > \max\{U,\ dDc + (K/d^m) c^n R^{m+n}\}. \tag{27}$$

If S is the steady slope in Theorem 4.1 with uplift $\overline{U}$, then

$$0 \leq -z_r(t, r) \leq S(r) \leq \frac{\overline{U}}{dD} r \qquad (t \geq 0). \tag{28}$$

Writing z_* for the steady state with the actual uplift U,

$$|z(t, r) - z_*(r)| \leq C_{\mathrm{in}} e^{-2dDt/R^2} (R^2 - r^2), \tag{29}$$

where $C_{\mathrm{in}} = \sup_{r<R} |z_{\mathrm{in}}(r) - z_*(r)|/(R^2 - r^2) < \infty$.

Proof. Local solution and positivity. Work first on the Cartesian ball, with $k(x) = K(|x|/d)^m$ and $F(x, p) = U - k(x)|p|^n$. Let $E = \{u \in C^1(\overline{B}_R) : u|_{\partial B_R} = 0\}$ with its C^1 norm. On each bounded subset of E, $F(x, \nabla u)$ is Lipschitz in u as a map into $C(\overline{B}_R)$, including $n = 1$. If $S_D(t)$ is the Dirichlet heat semigroup, its estimates

$$\|S_D(t) f\|_\infty \leq C\|f\|_\infty, \qquad \|\nabla S_D(t) f\|_\infty \leq C t^{-1/2} \|f\|_\infty \quad (0 < t \leq 1)$$

make the Duhamel map a contraction on a ball in $C([0, T]; E)$: its Lipschitz constant is at most $C_M(T + 2\sqrt{T})$. The homogeneous heat evolution is bounded in E for initial data in E. Thus the existence time is bounded below on bounded C^1 sets, and a finite maximal time requires an unbounded C^1 norm. This is the concrete heat-semigroup implementation of the local and continuation framework of [10, Section 3.3].

For clarity, regularity precedes differentiation of the slope equation. On a compact positive-time interval the bounded forcing and heat smoothing first give spatial and temporal Hölder regularity of ∇z with some exponent $\theta > 0$. Since k is $C^{0,\alpha}$ for $\alpha < \min(1, m)$ and $|p|^n$ is locally Lipschitz, the composed forcing is parabolically Hölder after reducing the exponent. Interior and Dirichlet boundary Schauder regularity then give $z \in C^{2+\theta,1+\theta/2}$ on smaller positive-time cylinders. The stated initial compatibility and $C^{2,\alpha}$ data give the corresponding regularity through $t = 0$ locally. In particular a bounded Hessian and radial symmetry imply $z_r(t, 0) = 0$ and $s(t, r) = O(r)$ uniformly on compact local-existence time intervals. After strict decrease is established, $s > 0$ on every closed annulus; there k and s^n are smooth and parabolic bootstrapping justifies the classical differentiated equation below. No differentiation of $|p|$ at $p = 0$ is used for $n = 1$. Positivity follows by the maximum principle, since the gradient term vanishes at an interior minimum and $U > 0$.

Preservation and strictness of radial decrease. Rotation invariance and uniqueness preserve radial symmetry. Reflect an outer spherical cap $\{x_1 > \lambda\} \cap B_R$, $\lambda > 0$, across $x_1 = \lambda$. The reflected point has smaller radius, so its k is no larger. The difference "inner reflected elevation minus outer elevation" satisfies a linear parabolic inequality with bounded drift and nonnegative forcing, obtained by the Lipschitz difference of the gradient nonlinearities. Its initial and parabolic boundary values are nonnegative. The maximum principle preserves their ordering. Strong comparison and the boundary point lemma on the reflection plane give $z_r < 0$ at every positive interior radius and positive time. The boundary point lemma for z gives the same conclusion at R. These arguments are applied on finite local-existence intervals, where the gradient is bounded.

A stationary upper envelope for the slope. Set $s = -z_r$. On $r > 0$, strict positivity permits differentiation, giving the important sign convention

$$s_t = D\left(s_{rr} + \frac{d-1}{r}s_r - \frac{d-1}{r^2}s\right) + k's^n + nks^{n-1}s_r. \tag{30}$$

Both nonlinear terms have a plus sign. A naive assertion that a positive interior slope maximum can never grow would therefore be false. At R, the fixed elevation yields

$$D\left(s_r + \frac{d-1}{R}s\right) + k(R)s^n = U. \tag{31}$$

The stationary slope S for $\overline{U}$ solves the same interior equation (30) and has the right side $\overline{U}$ in (31). Condition (27) and a first-crossing comparison with cr in its first-order ODE give $S \geq cr \geq s_{\text{in}}$.

Here is a center-safe comparison proof for $s \leq S$, including $m + n < 2$. On any compact local-existence time interval, $s = O(r)$, $S \asymp r$. The difference $e = s - S$ solves

$$e_t = D\mathcal{L}e + B(t, r)e_r + C(t, r)e, \quad \mathcal{L} = \partial_{rr} + \frac{d-1}{r}\partial_r - \frac{d-1}{r^2},$$

where, near zero,

$$|B(t, r)| \leq C_T r^{m+n-1}, \qquad |C(t, r)| \leq C_T r^{m+n-2}.$$

These bounds follow from secants of s^n and s^{n-1}; the latter secant is taken against the strictly positive $S \asymp r$. They remain valid for $1 < n < 2$. Indeed, write $s = ra$ and $S = rb$, where a is bounded and nonnegative and b is bounded above and below by positive constants. The divided difference of u^{n-1} between a and b is bounded on this set, even if $a = 0$. For $n = 1$ that divided difference is zero. At R, $De_r + \beta(t)e = U - \overline{U} < 0$, where $\beta(t) \geq 0$. Choose $0 < \gamma < 1$ and $\psi(r) = r^\gamma$. Then

$$\frac{D\mathcal{L}\psi + B\psi' + C\psi}{\psi} = \frac{D(\gamma - 1)(\gamma + d - 1)}{r^2} + O(r^{m+n-2})$$

is bounded above: its leading singular term is negative. Thus $\varepsilon e^{Mt}\psi$ is a strict supersolution for sufficiently large M, and it satisfies the positive Robin inequality at R. Since $e = O(r)$, $e/\psi \to 0$ at the center uniformly on the time interval. First-contact comparison therefore occurs neither at the center nor at the outer boundary, and the interior maximum principle gives $e \le \varepsilon e^{Mt}\psi$. Letting $\varepsilon \downarrow 0$ proves $s \le S$. Theorem 4.1 now gives (28).

Continuation. Integrating the bounded slope from R also bounds the elevation. The local semilinear continuation criterion excludes a finite maximal time. This proves global existence; uniqueness is inherited from local comparison.

Relaxation. Let $w = z - z_*$. Since both slopes are nonnegative,

$$w_t = D\Delta_r w + B_*(t,r)w_r, \qquad B_* = k(r)\frac{s^n - s_*^n}{s - s_*} \ge 0.$$

Use the continuous secant at equality. The coefficient is bounded by (28) and $n \ge 1$. For $\phi = R^2 - r^2$,

$$(D\Delta_r + B_*\partial_r)\phi = -2dD - 2B_* r \le -\frac{2dD}{R^2}\phi.$$

Comparison with $\pm C_{\mathrm{in}} e^{-2dDt/R^2}\phi$ proves (29). The boundary quotient in C_{in} is finite because the initial and steady elevations vanish at R and have bounded slopes. □

Remark 5.2 (Stability is radial only). The theorem does not suppress transverse channelization. The contributing area stays r/d because the flow geometry is held within the invariant radial class. The general nonradial evolution is not reduced to (26). Nor is the theorem stated for $0 < n < 1$, where the gradient Hamiltonian is not locally Lipschitz at zero. The displayed decay exponent tends to zero as $D \downarrow 0$; the estimate is not uniform in vanishing diffusion and does not justify exchanging the long-time and zero-diffusion limits.

Corollary 5.3 (Contraction between radial solutions)**.** *Let λ_1 and Φ be the first Dirichlet eigenvalue and positive radial eigenfunction of $-\Delta$ on B_R, with $\Phi(0) = 1$. For any two solutions of Theorem 5.1 with the same parameters,*

$$\sup_{0\le r<R}\frac{|z_1(t,r) - z_2(t,r)|}{\Phi(r)} \le e^{-D\lambda_1 t}\sup_{0\le r<R}\frac{|z_1(0,r) - z_2(0,r)|}{\Phi(r)}.$$

Taking $z_2 = z_*$ gives relaxation at rate $D\lambda_1 > 2dD/R^2$. The weighted initial constant is the displayed quotient, not C_{in} from the quadratic-weight estimate.

Proof. The difference satisfies $w_t = D\Delta_r w + Bw_r$, where $B = k(s_1^n - s_2^n)/(s_1 - s_2) \ge 0$ is bounded, using the continuous secant at equality. Near zero $B = O(r^{m+n-1}) \to 0$, so the Cartesian radial drift extends continuously at the centre. The eigenfunction equation gives $(r^{d-1}\Phi')' = -\lambda_1 r^{d-1}\Phi$, hence $\Phi' < 0$ for $r > 0$. Consequently $(D\Delta_r + B\partial_r)\Phi \le -D\lambda_1\Phi$. Comparison with both signs of the exponential barrier proves the estimate. The initial quotient is finite because Φ has a simple boundary zero and both elevations vanish there with bounded derivatives. To compare rates, put $T = (R^2 - r^2)/(2d)$. Green's identity gives $\int \Phi = \lambda_1 \int T\Phi < \lambda_1 R^2(2d)^{-1}\int \Phi$. This proves the strict inequality. This contraction remains radial only. □

6 Universal small-diffusion profile

Keep K, U, R, m, n fixed and let $D \downarrow 0$. Define

$$p = \frac{m}{n}, \quad \ell = \left(\frac{d^m D^n}{KU^{n-1}}\right)^{1/(m+n)}, \quad C_0 = d^{m/n}(U/K)^{1/n}. \tag{32}$$

Apart from the geometric factor $d^{m/(m+n)}$, the length scale is the known specific-area scale in [12, Eq. 17], after absorbing their grid-width factor into K. Dimensional analysis alone does not establish the profile, its convergence, or the constants below. Then $s_D(r) = (U\ell/D)y(r/\ell)$, where the single universal profile solves

$$y' + \frac{d-1}{\rho}y + \rho^m y^n = 1, \qquad y(\rho) \sim \rho/d \quad (\rho \downarrow 0). \tag{33}$$

It is the positive global solution already proved in Theorem 4.1.

Lemma 6.1 (Outer profile, including its first correction)**.** *For every $m, n > 0$ and $d \geq 1$,*

$$\rho^p y(\rho) = 1 - \frac{d-1-p}{n}\rho^{-(p+1)} + o(\rho^{-(p+1)}) \qquad (\rho \to \infty). \tag{34}$$

Proof. Put $w = \rho^p y$ and $c = d - 1 - p$. Direct substitution gives

$$w' = \rho^p(1 - w^n) - \frac{c}{\rho}w. \tag{35}$$

The bound $y \leq \rho/d$ implies $w/\rho \leq \rho^p/d$. If $c < 0$, consequently $w' \leq \rho^p(1 + |c|/d - w^n)$; for $c \geq 0$ the simpler bound applies. This gives a global upper bound for w. For any fixed $\varepsilon > 0$, the vector field at $1 - \varepsilon$ and $1 + \varepsilon$ points toward that interval once ρ is large. Outside the interval its magnitude is bounded below by a positive multiple of ρ^p, whose integral diverges. Thus the solution enters and stays in the interval, proving $w \to 1$.

For the correction put $a = -c/n$, $q = p + 1$, and consider $W_\pm = 1 + (a \pm \varepsilon)\rho^{-q}$. Their residuals in (35) have respectively the supersolution and subsolution signs, since the leading terms are $\pm n\varepsilon/\rho$, while their derivatives and the remaining terms are $O(\rho^{-p-2})$. Near one the right side of (35) is strictly decreasing in w. If w remained above W_+, its gap derivative would eventually be at most $-n\varepsilon/(2\rho)$, an impossibility after integration. The lower barrier is analogous. Both barriers are invariant after entry. Letting $\varepsilon \downarrow 0$ proves (34). □

Lemma 6.2 (Second outer correction)**.** *Put $p = m/n$, $q = p + 1$, $c = d - 1 - p$, and*

$$A_1 = -\frac{c}{n}, \qquad A_2 = \frac{c\{(3-n)c - 2q\}}{2n^2}.$$

Then the universal profile satisfies

$$\rho^p y(\rho) = 1 + A_1\rho^{-q} + A_2\rho^{-2q} + O(\rho^{-3q}).$$

The constants implicit in the remainder may depend on d, m, n. If $c = 0$, the stronger bound $1 - \rho^p y(\rho) = O(\exp[-k\rho^q/q])$ holds for every $0 < k < n$.

Proof. The previously established convergence $w = \rho^p y \to 1$ and equation $w' = F(\rho, w) = \rho^p(1 - w^n) - cw/\rho$ are used here. Set $P = 1 + A_1\rho^{-q} + A_2\rho^{-2q}$. Taylor expansion about one is legitimate for every real $n > 0$, since P is bounded away from zero for sufficiently large ρ. In the residual $\mathcal{R} = P' - F(\rho, P)$, cancellation of the coefficients of ρ^{-1} and ρ^{-q-1} gives

$$nA_1 + c = 0, \qquad nA_2 + \frac{n(n-1)}{2}A_1^2 + (c - q)A_1 = 0.$$

These identities give the displayed coefficients, and $\mathcal{R} = O(\rho^{-2q-1})$. For $e = w - P$, the mean-value theorem gives $e' = a(\rho)e - \mathcal{R}$, with $a(\rho) \leq -k_0\rho^p$ for some $k_0 > 0$ and all sufficiently large ρ. Variation of constants therefore bounds $|e(\rho)|$ by an exponentially small initial term and

$$C\int_{\rho_0}^{\rho} \exp\left[-k_0\int_t^{\rho} u^p\,du\right] t^{-2q-1}\,dt.$$

On $[\rho_0, \rho/2]$ the exponential makes the integral exponentially small. On $[\rho/2, \rho]$, bound t^{-2q-1} by $C\rho^{-2q-1}$ and the exponential by $\exp[-k_0(\rho/2)^p(\rho - t)]$. The remaining integral is $O(\rho^{-2q-1-p}) = O(\rho^{-3q})$. When $c = 0$, the exact equation $dw/d\tau = 1 - w^n$, with $\tau = \rho^q/q$ and $0 < w < 1$, gives $1 - w = O(e^{-k\tau})$ for every $k < n$ by the mean-value theorem near one. □

Theorem 6.3 (Sharp summit classification)**.** *For every fixed $r \in (0, R]$,*

$$s_D(r) \longrightarrow C_0 r^{-p}, \qquad z_D(r) \longrightarrow C_0 \int_r^R t^{-p}\,\mathrm{d}t.$$

At the summit there are exactly three regimes.

$$p < 1: \quad z_D(0) \longrightarrow \frac{C_0 R^{1-p}}{1-p}; \tag{36}$$

$$p = 1: \quad z_D(0) = C_0\{\log(R/\ell) + \mathcal{C}_{d,m,n} + o(1)\}; \tag{37}$$

$$p > 1: \quad z_D(0) \sim \frac{U\ell^2}{D}\,\mathcal{J}_{d,m,n}, \quad \mathcal{J}_{d,m,n} = \int_0^\infty y(\rho)\,\mathrm{d}\rho \in (0, \infty). \tag{38}$$

Here

$$\mathcal{C}_{d,m,n} = \lim_{S\to\infty}\left(\int_0^S y(\rho)\,\mathrm{d}\rho - \log S\right)$$

exists in the critical case. For $p < 1$, convergence of the elevation is uniform on the closed ball. For $p > 1$, the power of D in (38) *is $(n - m)/(m + n)$, explicitly*

$$\frac{U\ell^2}{D}\mathcal{J} = U\left(\frac{d^m}{KU^{n-1}}\right)^{2/(m+n)} \mathcal{J}\, D^{(n-m)/(m+n)}.$$

Proof. The identity $C_0 = U\ell^{p+1}/D$ and Lemma 6.1 give the pointwise slope limit. On compact intervals away from zero it is uniform, so it can be integrated. The exact summit identity is

$$z_D(0) = \frac{U\ell^2}{D}\int_0^{R/\ell} y(\rho)\,\mathrm{d}\rho. \tag{39}$$

When $p < 1$, the globally bounded $w = \rho^p y$ gives an integrable uniform bound $s_D(r) \le CC_0 r^{-p}$. Dominated convergence proves convergence of slopes in $L^1(0, R)$ and hence uniform convergence of elevations, including (36). At $p = 1$, (34) implies that $y - \rho^{-1}$ is integrable at infinity, giving (37). At $p > 1$, $y \sim \rho^{-p}$ is integrable at infinity, while $y \sim \rho/d$ is integrable at zero. Equation (39) proves (38). □

Corollary 6.4 (The finite-crest viscosity limit)**.** *For $0 < p < 1$, the continuous radial limit in Theorem 6.3 is a viscosity solution of*

$$K(r/d)^m|\nabla z|^n - U = 0$$

on the entire ball, with zero boundary values. For $p \ge 1$ the limiting profile is not a finite continuous viscosity solution on the closed ball because its summit diverges.

Proof. Away from zero the equation is classical. At zero the limit has the form $z(0) - Cr^{1-p}$, $C > 0$. No smooth function touches this cusp from below at zero, so the supersolution condition there is vacuous. Every upper test has Hamiltonian value $-U \le 0$, proving the subsolution condition. This statement concerns the scalar limiting stream-power equation, not the different three-field wet–dry system. □

6.1 All weighted relief moments

For $\beta > 0$ put $V_\beta(D) = \int_0^R r^{\beta-1} z_D(r)\,\mathrm{d}r$. Physical integrated elevation in d dimensions is $|\mathbb{S}^{d-1}|V_d$.

Theorem 6.5 (Moment thresholds)**.** *For every $\beta > 0$,*

$$V_\beta(D) = \frac{U\ell^{\beta+2}}{\beta D} \int_0^{R/\ell} \rho^\beta y(\rho)\,\mathrm{d}\rho. \tag{40}$$

Consequently

$$\begin{aligned}
p < \beta+1: &\quad V_\beta(D) \to \frac{C_0 R^{\beta+1-p}}{\beta(\beta+1-p)},\\
p = \beta+1: &\quad V_\beta(D) = \frac{C_0}{\beta}\{\log(R/\ell) + \mathcal{C}_\beta + o(1)\},\\
p > \beta+1: &\quad V_\beta(D) \sim \frac{U\ell^{\beta+2}}{\beta D}\mathcal{J}_\beta, \qquad \mathcal{J}_\beta = \int_0^\infty \rho^\beta y(\rho)\,\mathrm{d}\rho.
\end{aligned}$$

The critical constant is the corresponding convergent logarithmic renormalization. The supercritical power of D is $\{n(\beta+1)-m\}/(m+n)$. In particular, total landscape volume has its threshold at $m/n = d+1$, not at the summit threshold $m/n = 1$.

Proof. Tonelli's theorem gives $V_\beta = \beta^{-1}\int_0^R t^\beta s_D(t)\,\mathrm{d}t$, proving (40). The tail is $\rho^{\beta-p}(1 + O(\rho^{-p-1}))$ by Lemma 6.1. Integrating this tail, or using dominated convergence when it is integrable, gives the three alternatives and their constants. Substitution of (32) gives the exponent of D. □

Theorem 6.6 (Rates for finite relief moments)**.** *Let $\beta > 0$, $p < \beta+1$, and put $h = \beta+1-p > 0$, $q = p+1$, $A_1 = -(d-1-p)/n$. Write*

$$V_\beta^{(0)} = \frac{C_0 R^h}{\beta h}.$$

If $0 < h < q$ (equivalently $\beta < 2p$), the constant

$$H_\beta = \int_0^\infty \{\rho^\beta y(\rho) - \rho^{\beta-p}\}\,d\rho$$

is absolutely convergent, and

$$V_\beta(D) = V_\beta^{(0)} + \frac{C_0}{\beta}\left\{H_\beta \ell^h + \frac{A_1 R^{h-q}}{h-q}\ell^q + O(\ell^{2q})\right\}.$$

If $h = q$ (equivalently $\beta = 2p$), the finite constant

$$H_\beta = \lim_{S\to\infty}\left\{\int_0^S \rho^\beta y(\rho)\,d\rho - \frac{S^q}{q} - A_1 \log S\right\}$$

exists, and

$$V_\beta(D) = V_\beta^{(0)} + \frac{C_0}{\beta}\ell^q\{A_1\log(R/\ell) + H_\beta\} + O(\ell^{2q}).$$

If $h > q$ (equivalently $\beta > 2p$), then

$$V_\beta(D) = V_\beta^{(0)} + \frac{C_0 A_1 R^{h-q}}{\beta(h-q)}\ell^q + o(\ell^q).$$

All statements concern fixed positive R, K, U, m, n as $D \downarrow 0$. They remain valid when displayed coefficients vanish; no nonzero leading correction is asserted in that case.

Proof. The exact scaling formula is

$$V_\beta(D) = \frac{C_0}{\beta}\ell^h I_\beta(R/\ell), \qquad I_\beta(S) = \int_0^S \rho^\beta y(\rho)\, d\rho.$$

The outer expansion gives

$$I'_\beta(\rho) = \rho^{h-1} + A_1\rho^{h-q-1} + O(\rho^{h-2q-1}).$$

When $h < q$, the difference of the first two terms in the definition of H_β is integrable at infinity; it is integrable at zero because $h > 0$ and $y(\rho) = O(\rho)$. Integrating the tail from S to infinity gives $I_\beta(S) = S^h/h + H_\beta + A_1 S^{h-q}/(h-q) + O(S^{h-2q})$. When $h = q$, subtraction of A_1/ρ for large ρ leaves an integrable remainder, yielding $I_\beta(S) = S^q/q + A_1 \log S + H_\beta + O(S^{-q})$. When $h > q$, integration of the first correction gives $I_\beta(S) = S^h/h + A_1 S^{h-q}/(h-q) + o(S^{h-q})$. Substitution of $S = R/\ell$ proves the assertions. □

Since $\ell^{p+1} = d^{m/n}D/(K^{1/n}U^{(n-1)/n})$, these corrections have respectively fractional-power, $D\log(1/D)$, and D scales. The second threshold $\beta = 2p$ distinguishes a correction controlled by the summit region from one controlled by the outer profile; it differs from the finiteness threshold $p = \beta + 1$. Vanishing coefficients can suppress the displayed correction, so a universally sharp rate is not claimed.

6.2 Exactly one maximum slope

Proposition 6.7 (The radial transition scale)**.** *The universal slope y increases to a single nondegenerate maximum at $\rho_* \in (0,\infty)$ and decreases thereafter. If $R > \ell\rho_*$, the physical maximum occurs at $r_* = \ell\rho_*$ and has size $(U\ell/D)y(\rho_*)$. Otherwise the maximum on the finite hillslope occurs at the boundary.*

Proof. For $d = 1$ the nullcline is $H(\rho) = \rho^{-p}$, strictly decreasing. The solution starts below it and must cross it, since remaining below would make y increasing while $H \to 0$. At a crossing $(y-H)' = -H' > 0$, so there can be only one. For $d > 1$ the positive nullcline satisfies $\rho^m H^n + (d-1)H/\rho = 1$. Writing $h = H/\rho$ shows that h strictly decreases from $1/(d-1)$ to zero and that H' has the sign of $(d-1)(m+1)h - m$. Thus H has one strict maximum. Initially $y \sim \rho/d < H \sim \rho/(d-1)$. A first crossing cannot occur while H is increasing, since $(y-H)' = -H' < 0$ there. Nor can it occur at the strict maximum: at a tangency there $(y-H)'' = -H'' > 0$, contradicting negativity immediately before contact. A crossing must eventually occur because $H \to 0$. After that point $H' < 0$, so a reverse crossing is impossible. At this crossing $y'' < 0$, proving nondegeneracy. □

7 An integrable radial family and deterministic checks

If $p = d - 1$ and $d > 1$, the transformed equation (35) separates. With $\tau = \rho^d/d$,

$$\frac{\mathrm{d}w}{\mathrm{d}\tau} = 1 - w^n, \quad w(0) = 0, \qquad \int_0^{w(\rho)} \frac{\mathrm{d}u}{1-u^n} = \frac{\rho^d}{d}. \tag{41}$$

Here $0 < w < 1$. For $n = 1$, $m = d - 1$, and for $n = 2$, $m = 2(d-1)$, respectively,

$$y(\rho) = \rho^{-(d-1)}(1 - e^{-\rho^d/d}), \qquad y(\rho) = \rho^{-(d-1)}\tanh(\rho^d/d).$$

These follow by elementary integration of (41) and provide exact benchmarks throughout the domain, not only asymptotic slopes.

In the $d = 2, m = n = 1$ case,

$$\int_0^S y(\rho)\,\mathrm{d}\rho = \tfrac{1}{2}\{\gamma_E + \log(S^2/2) + E_1(S^2/2)\}, \qquad \mathcal{C} = \frac{\gamma_E - \log 2}{2}. \tag{42}$$

Here $E_1(x) = \int_x^\infty e^{-t}\,\mathrm{d}t/t$. Differentiation and the zero-limit at $S = 0$ verify the identity. For $d \geq 3, m = d - 1, n = 1$, integration by parts gives

$$\mathcal{J} = \frac{d^{2/d-1}\Gamma(2/d)}{d-2}.$$

More generally, for $0 \leq \beta < d - 2$,

$$\mathcal{J}_\beta = \frac{d^{(\beta+2)/d-1}\Gamma((\beta+2)/d)}{d-\beta-2};$$

at $\beta = d - 2$ the logarithmic constant is $(\gamma_E - \log d)/d$. These formulas are derived by integrating the displayed exponential profile.

The accompanying script checks the ellipse identities (19), verifies interior residuals for the companion wet–dry benchmarks, and integrates (33) for seven parameter triples. The original seven-case replay checks leading-order normalization and sampled slope bounds. The added `verify_extensions.py` instead integrates $v = \rho^{p+1}(\rho^p y - 1)$ using a cancellation-safe nonlinear remainder. The table reports $v(120)$ from the tighter of two tolerances. At $\rho = 30, 60, 120$, it asserts $|v - A_1| < 4|A_2|\rho^{-(p+1)} + 2 \times 10^{-6}$ and tolerance-refinement differences below 2×10^{-7}. These are declared finite-sample test tolerances, not rigorous error bounds. The observed second correction is logged as diagnostic only.

d	m	n	observed correction	prediction	refinement difference
2	0.5	1	-0.50038109	-0.50000000	0.0000000000
2	1	1	0.00000000	-0.00000000	0.0000000000
2	2	1	1.00000231	1.00000000	0.0000000000
3	0.6	1.2	-1.25011903	-1.25000000	0.0000000001
1	1	2	0.25016660	0.25000000	0.0000000000
2	0.4	0.6	-0.55595794	-0.55555556	0.0000000000
3	3	1	1.00000002	1.00000000	0.0000000000

The $n = 1$ solutions agree with an independent integrating-factor quadrature at the tested radii to maximum absolute error below 1.3×10^{-11}. The critical renormalized integral is checked at radius 30 against the exact constant with absolute tolerance 10^{-8}, matching (42). These are floating-point consistency checks, not interval certificates. The symbolic checks prove polynomial identities, not the compactness, regularity, or continuation theorems.

8 Interpretation, obstructions, and remaining scope

A simple obstruction explains why existence cannot be asserted for every prescribed topography. Suppose a C^1 landscape has a nonempty compact sublevel basin inside Ω, with positive rainfall and no allowed sink. Choose a nonnegative test function $\varphi = \eta(c - z)$ supported in that basin, with $\eta' \geq 0$. If $q = -a\nabla z/|\nabla z|$, $a \geq 0$, then

$$q \cdot \nabla\varphi = a\eta'(c - z)|\nabla z| \geq 0, \qquad \int \rho\varphi = -\int q \cdot \nabla\varphi \leq 0,$$

a contradiction. One may justify the C^1 test by approximation. Ponding, filling or a sink changes the model; it is not a solution of the unmodified conservation equation. This

conservation argument is consistent with the established critical-point discussion in [4] and is not claimed as a new physical observation. The small-erosion states avoid the obstruction because $-\Delta z > 0$.

The results close several precise subproblems: weak-erosion steady existence on general smooth domains with positive variable forcing; all-parameter regular radial steady existence; global radial relaxation for $n \geq 1$; and a complete radial small-diffusion classification, including weighted observables and the slope transition. The constructive theorem additionally gives local nonradial uniqueness and parameter stability under constant forcing, $m + n \geq 1$, and its stated geometric margin. No nonradial existence for arbitrary erosion intensity, unrestricted general-domain uniqueness, nonradial dynamical well-posedness, stability to channel-forming perturbations, or evolution for $0 < n < 1$ in the same classical class is proved. The separate three-field wet–dry problem and the Díaz et al. conjecture are addressed by exact benchmarks, a corrected profile problem and conditional compatibility obstructions in the companion document. The whole supplied Problem 5 must therefore remain marked *partially resolved*, not solved.

Reproducibility and disclosure. All model-specific arguments are displayed. Standard Dirichlet elliptic regularity, the compact-convex fixed-point theorem, elementary ODE theory, heat-semigroup local theory, and maximum principles are the external mathematical tools. Source code, software versions, raw numerical outputs, a claim ledger and a hash manifest accompany this manuscript. No real-world landscape data have been fitted. Mathematical existence is not validation of the model's sediment-coupling assumptions; see the distinct physical-model analysis in [12]. The general-domain theorem uses zero elevation on the whole boundary, not arbitrary mixed, single-outlet or periodic conditions. The supplied referee audit and the present additional reviews are AI-assisted assessments, not human peer-review acceptance or formal proof certificates.

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References

[1] Anand, S. K., Bertagni, M. B., Drivas, T. D., & Porporato, A. (2023). Self-similarity and vanishing diffusion in fluvial landscapes. *Proceedings of the National Academy of Sciences, 120*(51), e2302401120. https://doi.org/10.1073/pnas.2302401120. Author text: https://arxiv.org/abs/2401.04113.

[2] Binard, J., Degond, P., & Noble, P. (2024). Well-posedness and stability analysis of a landscape evolution model. *Journal of Nonlinear Science, 34*, 20. https://doi.org/10.1007/s00332-023-09997-9. Author text: https://arxiv.org/abs/2211.09629.

[3] Birnir, B., & Landry, T. B. (2025). Existence theorems for PDEs modeling erosion and the optimal transportation of sediment. *Analysis and Mathematical Physics, 15*, 107. https://doi.org/10.1007/s13324-025-01100-7.

[4] Bonetti, S., Bragg, A. D., & Porporato, A. (2018). On the theory of drainage area for regular and non-regular points. *Proceedings of the Royal Society A, 474*, 20170693. https://doi.org/10.1098/rspa.2017.0693.

[5] Bonetti, S., Hooshyar, M., Camporeale, C., & Porporato, A. (2020). Channelization cascade in landscape evolution. *Proceedings of the National Academy of Sciences, 117*, 1375–1382. https://doi.org/10.1073/pnas.1911817117. Author text: https://arxiv.org/abs/1812.03696.

[6] Briane, M. (2014). Isotropic realizability of electric fields around critical points. *Discrete and Continuous Dynamical Systems–B, 19*(2), 353–372. https://doi.org/10.3934/dcdsb.2014.19.353. Author text: https://arxiv.org/abs/1306.0236.

[7] Briane, M. (2018). Reconstruction of isotropic conductivities from non smooth electric fields. *ESAIM: Mathematical Modelling and Numerical Analysis, 52*(3), 1173–1193. https://doi.org/10.1051/m2an/2018013.

[8] Richter, G. R. (1981). An inverse problem for the steady state diffusion equation. *SIAM Journal on Applied Mathematics, 41*(2), 210–221. https://doi.org/10.1137/0141016. Full text inspected, especially pp. 213–219.

[9] Gilbarg, D., & Trudinger, N. S. (2001). *Elliptic partial differential equations of second order*. Springer. https://doi.org/10.1007/978-3-642-61798-0.

[10] Henry, D. (1981). *Geometric theory of semilinear parabolic equations*. Lecture Notes in Mathematics, 840. Springer. https://doi.org/10.1007/BFb0089647.

[11] Hooshyar, M., Anand, S. K., & Porporato, A. (2020). Variational analysis of landscape elevation and drainage networks. *Proceedings of the Royal Society A, 476*, 20190775. https://doi.org/10.1098/rspa.2019.0775. Author text: https://arxiv.org/abs/1911.03519.

[12] Litwin, D. G., Malatesta, L. C., & Sklar, L. S. (2025). Hillslope diffusion and channel steepness in landscape evolution models. *Earth Surface Dynamics, 13*, 277–293. https://doi.org/10.5194/esurf-13-277-2025.